

MATH 3551 A — Abstract Algebra I
Additional Practice Midterm 1 — Version A — SOLUTIONS

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Problem 1. (30 points) **True or false; prove or disprove.**

- (a) **TRUE.** The disjoint permutation $(1\ 2\ 3)(4\ 5\ 6\ 7)$ lies in S_7 and has order $\text{lcm}(3, 4) = 12$.
- (b) **FALSE.** A homomorphism of additive groups must send the identity to the identity, but $\varphi(\bar{0}) = \bar{1} \neq \bar{0}$.
- (c) **TRUE.** Taking $x = 6$ gives $8x = 48 \equiv 6 \pmod{14}$. Equivalently, $\text{gcd}(8, 14) = 2$ divides 6, the solvability criterion.
- (d) **TRUE.** This is $\langle \bar{3} \rangle$ in the additive group $\mathbb{Z}/12\mathbb{Z}$: it contains $\bar{0}$, is closed under addition, and each element's additive inverse is again in the displayed set.
- (e) **FALSE.** A transposition is determined by its two moved points, so there are $\binom{8}{2} = 28$, not 32, transpositions in S_8 .
- (f) **FALSE.** The units are $(\mathbb{Z}/12\mathbb{Z})^\times = \{\bar{1}, \bar{5}, \bar{7}, \bar{11}\}$. Each nonidentity element squares to $\bar{1}$, so there is no element of order 4 and the group of order 4 is not cyclic.

Grading note. Most of the credit is in the reason: award the three reason-points for a specific permutation, congruence solution, subgroup check, count, or order obstruction. A circled answer without a reason earns only two points. For (d), accepting the generator description $H = \langle \bar{3} \rangle$ is useful evidence of closure but does not require a separate subgroup test.

Problem 2. (25 points) **Arithmetic modulo 91 and 63.**

- (a) The Euclidean algorithm gives

$$91 = 2 \cdot 37 + 17, \quad 37 = 2 \cdot 17 + 3, \quad 17 = 5 \cdot 3 + 2, \quad 3 = 1 \cdot 2 + 1.$$

Substituting backward,

$$1 = 3 - 2 = 6 \cdot 3 - 17 = 6 \cdot 37 - 13 \cdot 17 = 32 \cdot 37 - 13 \cdot 91.$$

Thus one choice is $u = 32$ and $v = -13$.

- (b) The identity in (a) says that $\overline{32}$ is the inverse of $\overline{37}$ in $\mathbb{Z}/91\mathbb{Z}$. Multiplying the congruence by 32 gives

$$x \equiv 32 \cdot 14 = 448 \equiv 84 \pmod{91}.$$

- (c) Since $\gcd(21, 63) = 21$ divides 42, solutions exist. Dividing the congruence and modulus by 21 gives $x \equiv 2 \pmod{3}$. Thus the solution classes in $\mathbb{Z}/63\mathbb{Z}$ are

$$\overline{x} = \overline{2 + 3k}, \quad 0 \leq k < 21.$$

These are 21 distinct classes, as expected from the gcd.

- (d) Since $91 = 7 \cdot 13$ is a product of distinct primes,

$$|(\mathbb{Z}/91\mathbb{Z})^\times| = \varphi(91) = 91 \left(1 - \frac{1}{7}\right) \left(1 - \frac{1}{13}\right) = 72.$$

Grading note. In (a), the backward substitution is the method credit. In (b), the inverse must come from the Bézout identity; a correct residue without work earns only partial credit. In (c), award four points for the reduced congruence $x \equiv 2 \pmod{3}$ and four for the complete parametrization of the 21 classes. In (d), require either the Euler-product formula or an equivalent count excluding multiples of 7 and 13.

Problem 3. (25 points) **A matrix subgroup.**

- (a) Direct multiplication gives

$$A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I, \quad A^4 = I.$$

Since $A^2 \neq I$, $|A| = 4$.

- (b) Since $B^{-1} = B$,

$$BAB^{-1} = BAB = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = A^{-1}.$$

Also

$$AB = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (AB)^2 = I.$$

- (c) The relation in (b) is equivalent to $BA = A^{-1}B$; by induction, $BA^i = A^{-i}B$ for every integer i . In any word in A and B , use this relation to move every B to the right, and then use $B^2 = I$ and $A^4 = I$. Thus every element is A^i or A^iB with $0 \leq i < 4$. Explicitly these eight matrices are

$$I, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, -I, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

The four matrices A^i are distinct because A has order 4. The four matrices A^iB are distinct for the same reason, and no A^iB equals an A^j : their determinants are -1 and $+1$, respectively. Hence these are exactly eight distinct elements of H .

- (d) We have

$$AB = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = BA.$$

Thus H is nonabelian. Every cyclic group is abelian, so H is not cyclic.

Grading note. Part (c) is the proof portion: award most of its points for the rewriting argument, then for the eight correct matrices and the determinant argument proving distinctness. In (d), the displayed noncommuting pair is the decisive obstruction; merely saying that H “looks like” a dihedral group is not a justification.

Problem 4. (20 points) **Subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_4$.**

(a) The orders are

$$|(1, 0)| = 2, \quad |(0, 1)| = 4, \quad |(1, 1)| = 4, \quad |(1, 2)| = 2.$$

In a direct product the order is the least common multiple of the component orders.

(b) We have

$$\langle(1, 1)\rangle = \{(0, 0), (1, 1), (0, 2), (1, 3)\},$$

$$\langle(1, 2)\rangle = \{(0, 0), (1, 2)\},$$

$$\langle(0, 1)\rangle = \{(0, 0), (0, 1), (0, 2), (0, 3)\}.$$

(c) The set K contains the identity. It is closed under addition because it is precisely the set of pairs whose two coordinates have order dividing 2; explicitly, adding any two displayed pairs gives another displayed pair. Every element of K is its own inverse, so $K \leq G$. From the lists in (b),

$$K \cap \langle(1, 1)\rangle = \{(0, 0), (0, 2)\}, \quad K \cap \langle(0, 1)\rangle = \{(0, 0), (0, 2)\}.$$

(d) No element has order 8 (the largest possible order is 4), so the group of order 8 is not cyclic.

Grading note. In (a), the product-order/lcm observation earns the method credit. In (c), four points are for a valid subgroup proof and two points for each intersection. A list of K with no closure or subgroup-test justification does not earn the proof credit.